

DISASTER OPS-CENTER

Multi-Source Disaster Intelligence & Response Support System

Document Type:	Project Research & Technical Specification Dossier
Target Platform:	Responsive Web Application (SIH 2026 Student Prototype)
Primary Audience:	General Public & Emergency Coordinators in India
Authored By:	AI Coding Assistant Pair Program with Developer
Status:	Reverted Baseline Stable (17 Ingested Regions & Vernacular Media)
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Notice: This compilation dossier contains the combined product requirements, design guidelines, and architectural research conducted during the build of the SIH 2026 emergency intelligence web application prototype.

1. Product Vision & Requirements

During natural disasters like floods, cyclones, and landslides, information is often heavily fragmented across weather agencies, local municipal updates, regional television networks, social platforms, and citizen reports. The primary goal of this project is to centralize and filter these feeds into a location-aware, accessible dashboard.

1.1 Core Problems Addressed

- **Information Fragmentation:** Critical data sits across distinct, disconnected siloes.
- **Public Contradictions:** Unverified crowdsourced news often directly contradicts official advisories.
- **Accessibility Barriers:** Raw meteorological data is rarely interpreted in simple, actionable terms for lay citizens.
- **Alert Fatigue:** High-volume un-deduplicated notification flows cause users to ignore warnings.

1.2 The Three Core Questions

To provide effective decision support, the interface must answer three questions for any Indian citizen within 15 seconds:

UX Question	Technical Capability
What is happening?	Multi-source signal ingestion and real-time interactive GIS map layer visualization.
Can I trust it?	Visible verification weight scores, source type categorization, and corroboration statuses.
What should I do?	Rule-based emergency response recommendations mapped specifically to local disaster type and severity.

1.3 Critical Product Principle

The system is a decision-support platform, not an autonomous emergency command authority. Official government notifications (NDMA, SDMA) must always override system recommendation engines. No recommendation is ever presented as a guaranteed safety directive, and a permanent disclaimer is displayed across all screens.

2. Design Framework & Visual System

The interface adopts a high-contrast 'Tactical Ops-Center' theme, ensuring optimal visibility in low-light and emergency situations. It prioritizes clarity over visual density, storing raw statistics one layer below simple visual indicators.

2.1 Visual Tone & System States

A calm, authoritative visual tone is achieved using neutral backgrounds paired with a bold, functional brand blue. Gradients and glowing neon animations are intentionally excluded to keep the emphasis on readability. The application natively handles various offline, loading, and empty state representations without utilizing disruptive full-screen spinners.

2.2 Design Token & Color Palette Specification

Token Category	Value / Hex	Operational Usage
Primary Brand Blue	#123B5D	Headers, main titles, core navigation elements.
Action Accent Blue	#155A85	Interactive buttons, primary CTAs, alerts feed.
Low Risk Green	#18864B	Low-severity incidents and green verification indicators.
Moderate Risk Amber	#C88719	Moderate alerts, warnings, and stale indicator state.
High Risk Orange	#D65A1F	High-severity alerts, severe weather markers.
Critical Risk Red	#C62828	Extreme hazards, flash flood and tsunami alarms.
Dark Mode Neutral BG	#081425	Main screen container backdrop for Ops-Center theme.
Light Mode Neutral BG	#F7F9FB	Day/Light theme panel backgrounds.

2.3 Accessibility Constraints (WCAG 2.2)

To comply with accessibility standards, color is never used as the sole medium for conveying risk or severity. Every marker combines shape, typography (e.g., explicit labels like CRITICAL, HIGH), and icon symbols alongside its colors. Furthermore, target color ratios preserve contrast above 4.5:1 for normal text and 3:1 for graphical UI elements.

3. Datasource Ingestion & Regional Coverage

The prototype connects 10 distinct ingestion providers spanning 17 critical regions across the Indian subcontinent. These include major regional news wires and vernacular broadcast networks to maximize coverage.

3.1 Ingestion Sources Grid

Sectors / Regions	Primary Hazard	Ingested Source Feeds
Ahmedabad (Gujarat)	Urban Underpass Flooding	IMD Sabarmati radar, ABP Asmita, CWC telemetry
Wayanad (Kerala)	Slope Failures & Landslides	GSI slope telemetry, Asianet News Kerala
Puri (Odisha)	Cyclones & Wave Surges	INCOIS wave model, OTV Odisha, OSDMA feeds
Joshimath (Uttarakhand)	Cloudburst & Glacial Outflow	ISRO glacial radar array, SEOC Himachal/Uttarakhand
Guwahati (Assam)	Brahmaputra Flooding	CWC gauge telemetry, Pratidin Time Assam
Kolkata (West Bengal)	Estuarine Tidal Ingress	INCOIS coastal warning system, Telegraph Bengal
Mumbai (Maharashtra)	Monsoonal Inundation	MCGM automated weather array, local news tickers
Delhi NCR	Yamuna River Spillover	CWC old railway bridge gauge, Delhi disaster feeds

3.2 Multi-Tier Verification Logic

Disaster signals are corroborated across four distinct validation tiers in real time. Verification weights are computed as follows:

- 1. Meteorological Sensors & Satellites (Weight +40%):** Directly ingested telemetry from official government arrays.
- 2. Regional Media & Vernacular Tickers (Weight +30%):** Credible broadcast confirmations from state bureau vans at the scene.
- 3. Citizen Crowdsourcing (Weight +15%):** Public reports clustered geographically within a 2 km radius.
- 4. Official Agency Bulletins (Weight +15%):** NDMA, SDRF, or Municipal bulletins confirming active status.

4. Incident Lifecycle & Map Visibility Rules

A critical problem with disaster maps is clutter: historical or unverified reports remain visible, making zones appear more hazardous than they currently are. The system utilizes a strict state transition matrix to manage map visibility.

4.1 Lifecycle State Transitions

Reported incidents move through a clear lifecycle from creation to resolution or rejection:

[REPORTED] → [UNDER REVIEW] → [CORROBORATED] → [VERIFIED]

Alternatively, incidents can transition to [STALE] or be flagged as [REJECTED/FALSE].

4.2 Freshness Threshold Rules

If no corroborative updates are received within the configured freshness window, unverified incidents automatically decay to a lower-priority state:

Incident Status	Freshness Window	Lifecycle Behavior after Expiry
Reported	6 Hours	Transition to STALE. Marker opacity drops to 0.45. Excluded from active emergency counts.
Under Review	6 Hours	Transition to STALE. Marker opacity drops to 0.45. Requires re-corroboration.
Corroborated	12 Hours	Transition to STALE. Bullet styling switches to dotted circle/faded status.
Verified	No automatic expiry	Remains active on map indefinitely until resolved by an emergency coordinator.
Rejected	Immediate action	Hidden from public map views immediately. Retained in database for audit logs only.

4.3 Coordinator Control & Audit Trails

Authorized coordinators have access to administrative overrides. When an action is taken (Verify, Resolve, or Reject with mandatory reason), the system creates an immutable Audit Log entry recording the operator ID, timestamp, state change, and justification. This trail is displayed chronologically inside the event dossier.